

Effectiveness of VR Exercise at Helping Adults to Become Healthier

Inzamam Ul Haque

Department of Data Science, IT University of Copenhagen, Denmark

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Abstract

The purpose of this research is to understand how effectively virtual reality can help adults to become healthier.

Through the conduction of seven interviews and data scraping, we came to the conclusion that VR exercise can not make a great impact on global health crisis despite having the same benefits as some real-life workouts. But VR has the potential to be a great tool for exercise for which we may have to wait a few more years.

This research is beneficial for people cautious about their health and athletes who may find VR exercise useful. It is also beneficial for VR industry as they can get an insight of detailed user experience of their product which can be useful for development of more user-friendly products in the future.

Introduction

Virtual reality has plethora of applications and physical exercise is one of them. As the VR technology evolved over the years, the virtual environment became more immersive and realistic. Virtual reality headsets are becoming more user friendly, less bulky and there are more VR applications now than a few years back. In contrary to traditional gaming, virtual reality has enabled us to be more physically active while playing games. There are many VR games that require physical movements such as leaning left and right, dodging, bending backwards or forwards. Although virtual reality is now a known tool for physical exercise, there are many

skepticisms about its effectiveness. This research paper tries to answer if an individual chooses virtual reality instead of traditional exercises, whether it is going to be as good as doing exercises in the traditional way. To find the answer of the research question, first we determined whether or not exercising using VR gives the same result as exercising in regular way. Then we explored the limitation of VR exercise from users' point of view. By following this process, we reached a conclusion that takes both, scientific and social problems into account.

Background

Being healthy is directly connected to staying fit. Over the past decades, the effects of physical activity (PA) on individual's health have been well documented (Chou CH, Hawang CL, & Wu YT, 2011). Physical exercise enables us to stay fit. But many people cannot exercise regularly due to work, accessibility to exercise equipment or for various other reasons. Also, for many people exercise is monotonous and not enjoyable, thus they tend to abstain from it. Despite the well-known benefits of physical activity participation, according to the World Health Organization (WHO) approximately 25% of adults and 80% of adolescents around the world are physically inactive partly due to societal and lifestyle changes (WHO, 2018). Most people do not find exercise to be fun activity. On the other hand, VR has proven to be desirable and much more enjoyable. If exercising using virtual reality gives the same benefits as going to a gym or as other traditional methods of exercise, it may help a vast portion of the population to get fit and healthy. In recent years, VR exercise has been recognized as a new approach to promote physical activity and health behaviors (Ahn S.J & Fox J, 2017).

VR exercise is not different from other types of aerobic exercise (EDH, 2022). But in contrary to regular exercise, VR is not monotonous as logging miles on a treadmill. A computer simulated environment also removes a variety of social problems. Some people prefer to exercise alone, some prefer to exercise in a group. In a virtual environment, we can be in a group or we can be alone in accordance to our preference. VR exercise can be just as effective at increasing one's calorie burn rate as treadmill running or cycling, depending on the intensity of the game (Gomez, Bagley,

Bolter, & Lee, 2018). With the rapid development of VR technology, VR exercise has the potential to be a game changer in the public health and fitness sector.

VR interventions like immersive memory training and a three-dimensional kayaking exercise program significantly improved older adults' short- and long-term memory (Park J, Yim J, 2016). Anxiety and mental health issue are common among older adults. VR exercise programs can relieve feelings of anxiety and depression and increase enjoyment and daily energy levels (McDonough, et al., 2019).

Obesity poses a major public health challenge. Each year, obesity contributes to an estimated 112,000 preventable deaths (Office of the Surgeon General (US)). VR can be a useful tool against this. VR exercise may not be popular at the moment, but numerous researches has shown it's the effectiveness.

Method

The research question was answered using qualitative data consisting of interviews and online scrapping. Our approach was to understand the experience of interviewees. We determined that quantitative data would not be enough to answer our research question. The reasons behind VR exercise being effective or ineffective can be more clearly explored using this method. After finding out that VR exercise has almost the same benefits as some real-life workouts, we built our interview guide to explore the reasons behind VR exercise not being popular as conventional exercise.

Many scientific journals, personal blogs, public surveys are available on internet about VR and exercise in general and gaming focused exercises in specific. We determined that these could provide greater source of data than observation. We also felt that data scraping will be easier to perform as observation in a VR center was both out of the scope and budget of this research and also time consuming. We searched for relevant journals using Google Scholar. We also used news and magazine articles relevant to VR exercise, gaming and physical therapy.

To examine whether exercising using VR instead of tradition exercising methods can help an individual to do more exercise, interviews with seven different individuals were conducted. The interview was structured as a semi-structured interview with an interview guide. We decided to conduct interviews individually, by each researcher, differing in numbers based on how many subjects a researcher could find in the limited time frame. Some interviews were conducted at home to create a comfortable environment where the interviewee was treated like a house guest. In other cases, the researchers traveled to a location that was the most fitting and convenient for the interviewee thus trying to create the most comfortable environment for the subjects of the research. The interviewees were people of all social and economic backgrounds ranging from all ages, except children, with an emphasis on previous VR experience. Among the interviewees, five were male and two were female. Some of them had prior experience with VR and also used VR for exercise. One interviewee had no prior experience with VR. All interviewees were in between age 22 to 30 except one. While choosing candidates for interviewing, we focused on selecting people who are different from one another so that we can cover a broader group of people. We followed the interview guide most of the time, but also asked relevant follow-up questions when seemed necessary. However, all interviewees were asked whether VR exercise can help them to do more exercise and thus categorized them into two groups. One group who thought VR exercise can be an alternative to regular exercise, the other group thought the otherwise.

It is important to talk about the ethical consequence of the interviewees participating in the interview. The interviews were conducted after getting the interviewee's consent. Some of the interviews were recorded with the permission of the interviewees. It is important to mention that we did not have any written agreement with any interviewee. Some direct quotes from interviews are used in this paper. Although the data is not kept anonymously in a computer, no names are used in this research paper to avoid any unwanted complication and to protect the privacy of the interviewee.

We used YouTube videos as a source of data. Observing people exercising using VR was not possible. But it seemed important to us to get the feedbacks from people who are trying out VR to exercise. The authors' motivation behind making these videos were irrelevant to our research. But

their experience was relevant to answer our research questions. We chose to focus on the people who were playing VR games to exercise, not just to enjoy the game. We went through the comments on videos as they were a huge source of feedback from people all over the world. The comments are written by people who have done something actively to give feedback, whereas the people we interviewed, only gave feedback about VR exercise because we asked them to do so.

We used a mixture of both a priori and emergent coding. We did this because we had a mixture of different types of data and by using both coding methods, we attempted to decrease the bias of the researchers. We coded the interviews with regards to the meaning. The coding was done with the help of coding software (Miro) as well as manually. We stored the data in Google Document and Google Drive.

The research has followed an interactive model where the research question has changed several times during the research, depending on the findings of the data collection and analysis. It started with an interest in the potential of VR technology in helping people to do more exercise, thus formulating the first version of our research question which was “How can VR technology aid in people doing more exercise?”. Then we looked into sperate application created with the purpose of exercising in VR. While conducting the interviews we realized that we were unintentionally comparing VR exercise with real life exercises. This led to research question number two: “What potential does VR tech have in replacing means of traditional exercising?”. After conducting all the interviews and going through the gathered data, we arrived at our research question number three: “What kind of advantages and drawbacks stem from the use of VR technology as a form of exercise?”. After coding the interviews and comparing it with other data gathered from different sources, we saw it fit to alter our starting main research question, to emphasize the aim of the study more clearly, we arrived at the main question: “How effective can VR technology be at Helping adults become healthier?”

Findings

For people who has full time job, it is not always possible to go the gym or to exercise regularly. Also, most exercises are deemed to be boring. One interviewee exercised regularly by biking. When we asked that interviewee if she found biking entertaining, she said:

“Honestly, not very. But I have to get some kind of exercise. I don't want to get fat.”

The major motivation to exercise for most interviewees was physical fitness, specifically to avoid becoming overweight. There was a huge variety of motivating factors ranging from pure willpower to entertainment during exercise. One interviewee said:

“I enjoyed weight lifting. It was entertaining so it was not a problem. But when it came to running, I knew I just had to use my willpower and put up with it.”

There were also a variety of experiences with VR technology, but most concluded that it was a great cardiovascular exercise with lots of dodging movements and sweating, but that VR is not a reliable mechanism for muscle building. There is also a challenge with simulating exercises like weightlifting and swimming. Exercises that are easier to simulate will still require exercise equipment to be used with the VR system. Also, exposure to a virtual environment may cause symptoms that are similar to motion sickness symptoms. It is called virtual reality sickness and it is closely related to simulator and motion sickness. One interviewee experienced this issue when he tried virtual reality headset. He said:

“The flight simulator I used it for turned out to be very realistic and it seems my inner ear and my eyes are not connected in the right way so I get quite a lot of motion sickness from it.”

There have been many recent developments regarding VR technology. But VR headsets are still expensive. To use a VR headset, we also need a decent computer with a good dedicated graphics processing unit. Thus, virtual reality is not yet accessible for most people. Majority of the interviewees who own VR headsets, never used it for exercise. They only used virtual reality for gaming. When asked if VR exercise is as good as regular exercise, we observed a mixed reaction from the interviewees. Three interviewees out of seven (see tab. 2) thinks VR exercise can be a good substitution of traditional exercises. Three interviewees did not think that VR can be used for exercise and one interviewee was not sure about it.

Interviewee	Owns a VR Headset	Exercises Regularly (Traditional)	Experience with VR	Thinks VR exercise is as good as regular exercise
1	No	Yes	Negative	No
2	No	Yes	Negative	Not sure
3	Yes	Yes	Positive	Yes
4	No	No	Positive	Yes
5	Yes	Yes	Positive	Yes
6	Yes	Yes	Positive	No
7	Yes	No	Positive	No

Table 1: Interviewee given information

There are many VR games which focuses on physical movement of the body. We focused on VR games that are popular. Many games turned out to be as effective as regular exercises according to their MET score (see tab 1). Supernatural and Thrill of The Fight, both of these games were about equivalent to riding a bike. FitXR and Until You Fall were almost equivalent to playing tennis. Virtual Reality Institute of Health and Exercise at San Francisco State University measured the metabolic equivalent of task of these four games: Supernatural, Thrill of the fight, FitFR, Until You Fall. They came to a conclusion that playing these four games is almost the same doing certain exercise like biking, playing tennis or running. It gave us a viewpoint that some of our respondents lacked, as we had many weightlifters amongst our respondents. Although for some games the MET score is the same as exercises like bike riding, it does not mean they both has the same effect on

the body. Furthermore, what will be the result of playing these games regularly for exercise and how will it effect our body is still a topic that remained unexplored.

Activity	Metabolic Equivalent of Task (MET)
Supernatural (VR Game)	11.44
Thrill of the Fight (VR Game)	9.28
FitXR (VR Game)	7.94
Until You Fall (VR Game)	6.5
Bicycling	8.0
Tennis	7.3
Jogging	7.0
Walking	3.0
Sitting	1.0

Table 2: MET Score of different activities

According to data VR exercise is as good as real-life exercise. But when the interviewees were asked, most of them had a conflicting opinion to this fact. This enabled the interviewers to explore other the reasons behind VR exercise being less effective in helping adults to become healthier. To use VR for exercise, a minimum amount of open space is needed. One interviewee stated that he had to beware of their surroundings all the time and was always worried about bumping into things.

Discussion

We only interviewed a small amount of people. We could not interview anyone who uses VR for exercise. If we could observe people trying VR exercise for the first time and then interview them, it would have given us more valuable insights about their experience. Interviewees with previous virtual reality experience had provided more valuable feedback than those who had never used virtual reality. Therefore we tried to select interviewee with prior VR experience. Therefore a huge portion of the population who had never used VR were left out.

The questions that we had planned in our interview guide served as a great basis for the structure of the interview. The questions were structured in a way that it guided researchers in the direction of getting valuable feedback for answering our research question. But we got some feedbacks from the interviewees that some questions felt repetitive. While conducting an interview, the researchers also asked some extra questions whenever they felt that it could lead to valuable result. This semi-structured interview has allowed for the gathering of a higher quantity of relevant data for the research.

We were careful about our biases throughout the whole process of the research. Before making the interview guide, we identified our biases. We structured the interview guide in such a manner that the interview guide has enabled researchers to account for much of their biases, which could have hindered the value of the responses. But before conducting the interviews, we agreed to only interview people with previous VR experience. Again, those who had previous experience with VR, either already owned a VR headset or knew someone who had a VR headset that they could try. These factors largely diminished our sample base. Although other factors like staying healthy and making exercise more fun are common concepts for all groups of people.

Most of our interviewees were 25 years old or older which implies further bias in age of the interviewed people. It is logical to assume that younger people tend to use VR than older age group. As most of our interviewees were young, their opinions are expected to be over stated.

We also selected a video while data scraping titled “Virtual workout, real sweat: VR fitness apps, scientifically tested” from Washington Post. It provided a deep insight about gaming using VR,

especially about the games that can be considered as physical exercise. The narrator was interested in finding whether virtual reality gives as good a workout as taking a jog around the block or riding a bike. This was in direct alignment with our research question. The games that were tested in this video were scientifically proven to be as good as any physical exercise like bike riding, running or playing tennis. However, many questions remained unanswered in this video. For example, how long the narrator used VR for exercising or long-term effects of VR exercise. No negative effects of VR exercise were discussed in the video. There may also be a flawed implementation, as biking is a much different exercise than any of the games played on the VR headsets. Furthermore, no information was provided about the narrator's heartrate and calorie usage during exercise using virtual reality. This information was relevant for our research as many of our interviewees were concerned about obesity.

It is a fact that VR games can be as good as some real-life exercise. It is also true that VR exercise cannot be an alternative to some traditional exercises such as swimming or weightlifting. VR headsets are still expensive and many people do not use VR. While VR can be used for exercise, the technology is not yet developed enough and not accessible to everyone. Not exercising enough or not being healthy is an issue for people from all walks of life. VR exercise alone cannot solve this widespread problem of public health.

The use of VR in medical sector is very common nowadays. Although VR was primarily used for gaming in the beginning, now it is widely used for activities such as training pilots, astronauts, soldiers, to design something, for entertainment purposes, to treat PTSD (post traumatic stress disorder) and so on. Virtual reality has become a huge industry. On the other hand, fitness industry is at its peak with the market size of \$87.23 billion in 2019 (Stasha, 2022). As VR exercise is more entertaining than regular exercise, it has the potential to tap into the fitness industry. This can be a huge motivation to develop VR headsets suitable for exercise.

One of our assumptions was that exercise makes all people healthy, which is not true. Many people have certain medical condition for whom exercise or using VR may not be healthy at all.

This research is focused on effectiveness of VR exercise. One of the most serious ethical questions that has to be mentioned is that, "Are we getting detached from real world day by day?". New

diseases like nomophobia (Jenna, 2022) and SMD (Social Media Disorder) are the result of overusing smart devices. People are easily addicted to gaming. VR makes gaming more immersive and sometimes invokes psychological reaction same as real life situation. For example, the virtual blind date, which according to an experimental study provides a very real sense of nervousness, is one of several scenarios that psychologists have developed for training people to deal with challenging social situations (Hartanto, et al., 2014). It begs the question, “Is VR exercise good for our mental health?”

Conclusion

In this study, we asked the question: “How effective can VR technology be at helping adults become healthier?”. To answer the research question, we used qualitative data in the form of interviews and online scrape. Analysis of the data showed that VR can be as effective as some conventional exercises. But the drawbacks and limitation of VR technology makes it unsuitable for being used as an exercise tool for the general mass. At the moment, the main drawback of VR technology seems to be the price of the systems, which currently prevents any but the most dedicated athletes and gamers from having access to the benefits of VR exercise. For this reason, while it is an excellent motivator for exercise that can benefit a large group of people, the technology is not yet widely available enough to have an impact on the global health crisis.

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Appendix

Interview guide

- Does VR increase motivation in engaging in exercises/sport activities?

Name/Gender/Age

- Do you do exercise regularly?

IF YES

- What kind of exercise do you do?
- What motivated you to start that activity?
- How often do you exercise and do you enjoy it?
- How did you manage to keep up with the activity for a certain amount of time?
- How entertaining have you found that sport/activity?

IF NO

- Do you remember the last time you engaged in physical activity on a regular basis (maybe for a few months in a row)?
- How did you manage to keep up with the activity for a certain amount of time?
- Did you quit after a while and do you remember why?
- Do you think that the sport/activity was not entertaining?

BOTH

- Have you ever tried using VR headsets for doing an exercise?
- If so, what was your experience?
- If not, why not?
- What do you define as enjoyable physical activity?
- What perspective do you have for sport – do you believe sport is suffering?
- Would you prefer to exercise at home or outside? Why?
- Do you think you do not have enough time for training?
- Does being sociable, maybe part of a team, increase your motivation to engage in physical activity?

- Do you believe that training with VR headsets will decrease that satisfaction of being part of a team?
- Do you think training with VR headsets can help you have more spare time compared to regular training?

Themes and most common keywords from interview coding

- Preferred activity: Weightlifting, Calisthenics, Running
- Frequency: No regular exercise, exercises regularly, twice a week, five times a week
- Reasons for stopping: Moving, boring
- Viewpoint on sports: not entertaining, enjoyable, challenging, boring
- Motivating factors: fitness, fat, confidence, music
- Challenges: Time, safety
- Teams: prefers alone, team not important, enjoys community feel
- Preferred location: Gym, home, outside
- VR experience: Used for exercise, not used for exercise, VR for gaming
- VR opinions: Good for cardio, bad for muscle building, not practical
- VR concerns: Awareness of surroundings, motion sickness, isolation, sweating and headsets, limited sports
- VR movements: Dodging, upper body movements
- VR benefits: Increases satisfaction, no travel time, enjoyable

Interview Coding

- “I think the technology has promise, but it's not ready yet. I'm worried about neck strain from wearing a heavy headset for a long time, as well as motion sickness from a badly implemented interface. Not to mention that I don't see how some sports can be simulated with a headset, like swimming.”
- “I have to get some kind of exercise. I don't want to get fat.”
- “The flight simulator I used it for turned out to be very realistic and it seems my inner ear and my eyes are not connected in the right way so I get quite a lot of motion sickness from it.”
- “It shows a track, an imaginary track, say and it's easier to get oneself to work harder when you're going uphill or to catch the dinosaur and silly things. The whole team has to pedal harder to catch the dinosaur. So that's motivation, yes.”
- “I work out 5 times a week at the gym.”
- Yes. I own a VR console. Not exactly exercise. But there was that game super hot. After 5 minutes im all sweats to my a**. It is super fun. You also had to move a lot while playing Half Life: Alyx.
- “Well while playing you had to do dodging movements. Leaning left and right. You had to bend backwards and also do the whole thing in slow motion like in matrix.”
- “Outside in the gym. Working out at home did not give me enough muscle. It wasn't hard to maintain motivation. I kept doing exercises.”
- “I do not think VR could replace it. It isn't good for weight training.”
- “I would go to the gym. Weightlifting. And then I plan to move on to calisthenics.”

- “Yea I got really sweaty by the end. I played for like 30-40mins. It was a lot of fun of shooting and dodging.”
- “Outside. Doing it at home felt more difficult. I really don’t like it. I need the mindset of the gym otherwise I don’t have the motivation. My mind wanders off and I start thinking about going for a cigarette or something else instead of being focused.”
- “It is not equal to a muscle building exercise, but it can be a fun cardio experience. You are definitely doing more exercise than sitting in a chair. There is also this tech called catwalk where you can walk on a platform too. You have to play VR for a considerable time for it to be a good workout, I think. It needs new technology for it to be a proper tiring workout. The problem is that you move with the console.”
- “I have a full-time job. Also, I hang out with friends a lot after work. Therefore, I don’t have enough time for training or exercise.”
- “I don’t have to drive to certain place to exercise using VR. Can do that at home. Except that, the actual exercise takes the same amount of time.”
- “I wanted to be more physically active and basically I was feeling like I was 50 when I was 25.”
- “I love weight lifting, but I do not like running and I exercise 5 times a week (either weighting lifts or running). I also walk 5 km each day.”
- “It was not like the real experience of dancing and I felt like I could not move freely enough.”
- “Not really understanding your surroundings felt like it was ruining the experience, but maybe that was just this game.”
- “If VR technology becomes good enough maybe it will be useful if people are not in the same location - like different cities or countries and they train virtually and they have virtual competitions and stuff like that. But for traditional training- like being with your friends in the gym, I believe with VR headsets it will be worse and quite limiting at what you do. If you want to play volleyball with a VR headset, obviously you are kind of blind and you can’t do it. I believe no matter how good the technology is your senses are kind of dimmed.”

Video Source:

Link: <https://youtu.be/291LZGxZS5Q?list=PLElNMLD5xG9yTrSg8bvZFnLdlu2vsqMiE> - Scientific test (Washington Post)